



Parul University
Faculty of Engineering and Technology
Parul Institute of Engineering and Technology
Department of Artificial Intelligence and Data Science

Subject Name	Theory of Computations	A.Y	2026-27	
Subject Code	303105306	Semester	5	
Assignment-1				
Sr No	Question	COs	B.T	Competence
1	Define an alphabet, string, and language with suitable examples. Explain the concept of Kleene closure $(\Sigma^*)(\Sigma^*)(\Sigma^*)$.	1	2	Understand
2	What is a grammar? Explain the components of a grammar $G=(V,\Sigma,R,S)$ with an example.	1	2	Understand
3	Differentiate between terminal symbols and non-terminal symbols used in formal grammars.	1	4	Analyze
4	Explain the concepts of production rules and derivation in formal languages with suitable examples.	3	3	Apply
5	Describe the types of grammars in the Chomsky hierarchy. Compare Type-0, Type-1, Type-2, and Type-3 grammars.	3	4	Analyze
6	Explain the Chomsky hierarchy of languages and discuss the relationship between grammars and automata.	2, 3	2	Understand